Order Flow Imbalance — Conceptual Questions

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1. Why look at several book levels?

Watching only the best bid and ask shows what is happening *right now*, but not what sits one or two ticks away. A large buy order may be split and parked at levels 2 and 3 to avoid moving the quote. Tracking OFI for the first L levels,

$$\mathrm{OFI}_t^{(L)} = \sum_{\ell=1}^{L} (\Delta q_{\ell,t}^{\mathrm{bid}} - \Delta q_{\ell,t}^{\mathrm{ask}}),$$

captures that hidden pressure. Example: level 1 stays flat, but bid level 2 grows by +10,000 shares while the ask side is unchanged. The best-level OFI is zero, yet $\mathrm{OFI}^{(2)}$ jumps by +10,000, flagging buy interest that is not visible at the top of book.

2. Why choose Lasso over OLS for cross-impact?

Modelling each stock's return with the OFIs of many others yields a high-dimensional, collinear problem. OLS fits a coefficient for every link and overfits easily. Lasso adds an ℓ_1 penalty,

$$\min_{\beta} \sum_t (r_{i,t} - \beta^\top \mathrm{OFI}_t)^2 + \lambda \|\beta\|_1,$$

which drives most coefficients to zero. The result is a sparse, easy-to-interpret matrix of the few stocks that truly influence each other and a model that generalises better out-of-sample.

3. Why does OFI beat raw volume for short-term forecasts?

Volume counts trades but ignores direction; the same volume can arise from buyers lifting offers or sellers hitting bids. OFI adds the missing sign: positive when buy pressure dominates, negative when sell pressure dominates. The referenced paper shows OFI explains more one-minute return variance than volume because it separates buying from selling instead of mixing them.

References

[1] R. Cont, M. Cucuringu, and C. Zhang. *Cross-Impact of Order Flow Imbalance in Equity Markets. Quantitative Finance*, 23(10):1373–1393, 2023.